

Wachirawit Kaewdang

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SUMMARY

Computer Science graduate (Software Engineering track) from Thammasat University. I work across machine learning for time series – with a particular focus on model explainability rather than accuracy alone – and full-stack web development in React and Node.js. Seeking Machine Learning Engineer or Software Engineer roles.

EDUCATION

Thammasat University

2022 – May 2026

BSc Computer Science, Software Engineering track – Faculty of Science and Technology

PROJECTS

Stock Price Direction Prediction – LSTM + Self-Attention with Explainable AI

Senior project (CS403)

github.com/wachirawit-kaewdang/Project-CS403

- Built a deep learning pipeline predicting 3-day price direction for 10 US stocks across 5 sectors (2020–2025), from 27 engineered features: momentum and volatility, moving averages, RSI / MACD / Stochastic, volume and OBV, S&P 500 market context and bond yield.
- Architecture: LSTM with self-attention (3 layers, hidden size 128), trained with Focal Loss to handle class imbalance. **Average F1 0.710; best 0.754 (JPM).**
- Benchmarked against Random Forest and Logistic Regression baselines, added a VIX market-fear feature set, and built an LSTM + RF + LR majority-vote ensemble as the final best model.
- Applied SHAP (KernelExplainer) for per-prediction attribution and benchmarked it against Pearson correlation, with rank-disagreement and sector-level analysis – testing whether the model's reasoning matched conventional financial intuition.
- Evaluated quarterly fundamental data (P/E, EPS, market cap) and deliberately excluded it from training after finding that constant fill-forward injected noise into the daily series; retained as descriptive analysis and documented as a limitation.
- My contribution: data collection and preparation, and design of the training approach including model architecture. Two-person project.

Python • PyTorch • SHAP • scikit-learn • pandas • NumPy • Jupyter

Recipe Web Application – Full-Stack

CS369, group of 5

- React + Vite frontend with client-side routing, connected to a Node.js / Express REST API.
- My contribution: the MyRecipe management feature, recipe deletion, and integration of the frontend with the backend API.

React • Vite • JavaScript • Node.js • Express • REST APIs

HTTP Web Server from Scratch

CS322 Computer Networks

- Implemented an HTTP/1.0 server in Python using raw socket programming – no frameworks – serving static HTML and image content with correct status codes and content-type headers.

Python • sockets • TCP/IP • HTTP

TECHNICAL SKILLS

Languages	Python, JavaScript, Java, C, C++, SQL
ML & Data	PyTorch, scikit-learn, SHAP, pandas, NumPy, Jupyter – time series, feature engineering, explainable AI
Web	React, Vite, Node.js, Express, Django, REST API design, HTML/CSS
Databases	MySQL, MongoDB
Tools	Git, Docker, automated testing (TDD), object-oriented design

LANGUAGES

Thai	Native
English	Professional working proficiency – technical reading and writing